

Practical No. 2

AIM :- Program to create a list-based stack and perform various stack operations.

CODE -

#AIM :- To implement the stack using list

```
stack = []
```

```
while True:
```

```
    print("\n-----Stack Operations")
```

```
    print("1.Push")
```

```
    print("2.Pop")
```

```
    print("3.Peek")
```

```
    print("4.Check if empty")
```

```
    print("5.Size of stack")
```

```
    print("6.Exit")
```

```
choice = int(input("Enter the choice : "))
```

```
if choice == 1:
```

```
    size = int(input("How many element you want element : "))
```

```
    for i in range(size):
```

```
        element = int(input(f"Enter the {i} element : "))
```

```
        stack.append(element)
```

```
print("Stack : ", stack)
```

```
elif choice == 2:
```

```
    popelement = stack.pop()
```

```
    print("Stack after popping item : ", popelement)
```

```
elif choice == 3:
```

```
    topelement = stack[-1]
```

```
    print("Top element in stack : ", topelement)
```

```
elif choice == 4:
```

```
    if stack:
```

```
        print("Stack is not empty")
```

```
    else:
```

```
        print("Stack is empty")
```

```
elif choice == 5:
```

```
    size_of_stack = len(stack)
```

```
    print("Size of stack : ", size_of_stack)
```

```
elif choice == 6:
```

```
    print("Goodbye!")
```

```
    break
```

else:

print("Enter a valid output")

cont_choice = input("Do you want to continue(y/n) : ").lower()

if cont_choice != 'y':

print("Goodbye!")

break

OUTPUT –

```
-----Stack Operations
1.Push
2.Pop
3.Peek
4.Check if empty
5.Size of stack
6.Exit
Enter the choice : 1
How many element you want element : 4
Enter the 0 element : 9
Enter the 1 element : 6
Enter the 2 element : 8
Enter the 3 element : 4
Stack :  [9, 6, 8, 4]
Do you want to continue(y/n) : y

-----Stack Operations
1.Push
2.Pop
3.Peek
4.Check if empty
5.Size of stack
6.Exit
Enter the choice : 2
Poped item from stack :  4
Stack after popping item:  [9, 6, 8]
Do you want to continue(y/n) : y

-----Stack Operations
1.Push
2.Pop
3.Peek
4.Check if empty
5.Size of stack
6.Exit
Enter the choice : 3
```

Top element in stack : 8
Do you want to continue(y/n) : y

-----Stack Operations

- 1.Push
- 2.Pop
- 3.Peek
- 4.Check if empty
- 5.Size of stack
- 6.Exit

Enter the choice : 4
Stack is not empty
Do you want to continue(y/n) : y

-----Stack Operations

- 1.Push
- 2.Pop
- 3.Peek
- 4.Check if empty
- 5.Size of stack
- 6.Exit

Enter the choice : 5
Size of stack : 3
Do you want to continue(y/n) : y

-----Stack Operations

- 1.Push
- 2.Pop
- 3.Peek
- 4.Check if empty
- 5.Size of stack
- 6.Exit

Enter the choice : 6
Goodbye!